Expense Tracker - Technical Specification & Architecture Document

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1. Executive Summary & Architecture Overview

1.1 Executive Brief

A lightweight personal expense tracking application designed for single-user local use. The system enables the creation, listing, and deletion of expenses with a focus on immediate data persistence via browser localStorage. It features a summary dashboard providing real-time aggregate totals and spending history without the overhead of user authentication or a backend infrastructure.

1.2 Maturity Assessment

The specification is highly stable and structurally sound, with a near-perfect health index. While a formal 'Scope & Out-of-Scope' section is missing, the boundaries are clearly defined within the functional requirements. The project is READY for execution.

1.3 Technical Stack

• **Storage**: localStorage API

• **Data Format**: JSON (via localStorage)

• **Date Standard**: ISO 8601

1.4 Architectural Constraints

• Amount must be a positive number strictly greater than zero.

• Expenses must be sorted by date in descending order (newest first).

• Dashboard default view limit: exactly 50 most recent expenses.

• Future dates are permissible but must trigger a system warning.

• User authentication and multi-user support are strictly forbidden (Out of Scope).

• Editing or updating existing expenses is strictly forbidden in v1.

• Performance target: expense addition must be completed in under 30 seconds.

1.5 Critical Dependencies

• Browser localStorage API for data persistence and retrieval.

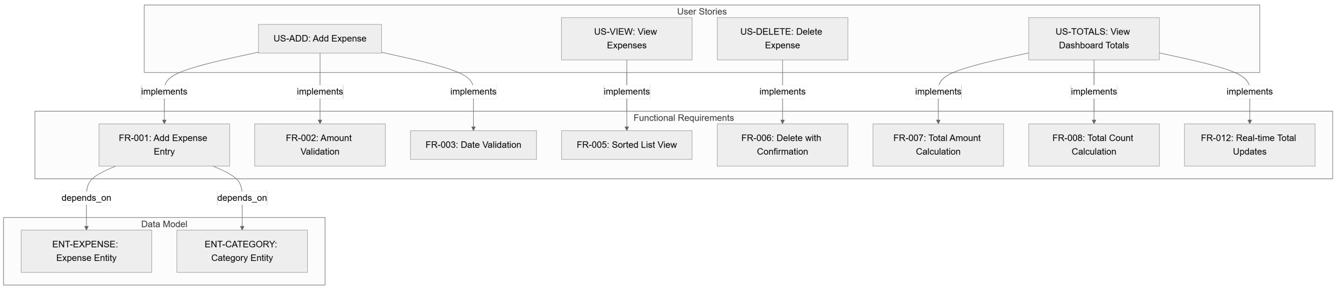
• Referential dependency between Expense entity and the predefined Category list.

• Real-time synchronization between the Expense data store and Dashboard aggregate totals.

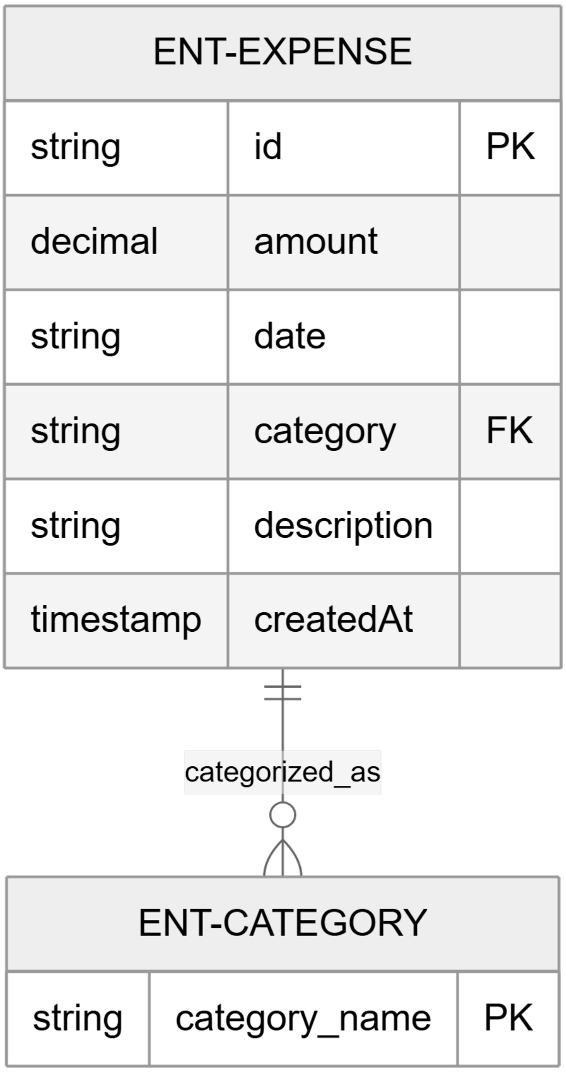
• Strict data integrity: Expense amount must be decimal and date must follow ISO format.

2. Architecture Workflows & Visual Diagrams

2.1 Requirements Traceability Matrix



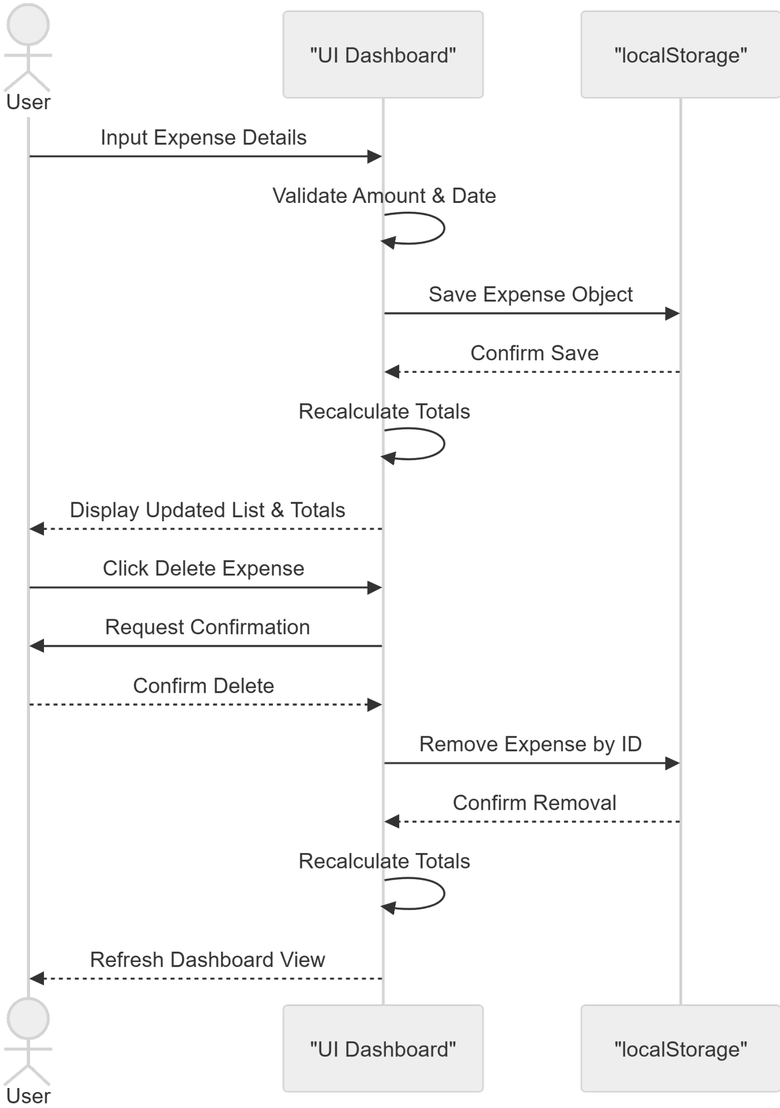
2.2 Expense Data Model



2.3 Add Expense Workflow

[Sch�ma non compil� : Diagramme #3]

2.4 Expense Management Sequence



3. Detailed Technical Specifications & Business Rules

3.1 Requirements Traceability

ID	Type	Description	Source / Relation
US-ADD	User Story	Add a new expense with amount, date, category, and description.	P1
US-VIEW	User Story	View a list of recent expenses to see spending history.	P1
US-DELETE	User Story	Delete an expense to correct mistakes or remove entries.	P1
US-TOTALS	User Story	See basic totals (total spent, count of expenses) on dashboard.	P2
FR-001	Functional	System MUST allow users to add a new expense (amount, date, category, description).	Implements US-ADD
FR-002	Functional	System MUST validate that amount is a positive number greater than zero.	Implements US-ADD
FR-003	Functional	System MUST validate date; future dates trigger warning but allow saving.	Implements US-ADD
FR-004	Functional	System MUST provide a predefined list of categories (Food, Transport, etc.).	Depends on ENT-CATEGORY
FR-005	Functional	System MUST display expenses sorted by date descending.	Implements US-VIEW
FR-006	Functional	System MUST allow users to delete an expense with confirmation.	Implements US-DELETE
FR-007	Functional	System MUST display total amount spent across all expenses.	Implements US-TOTALS
FR-008	Functional	System MUST display count of total expenses.	Implements US-TOTALS
FR-009	Functional	System MUST persist expenses to localStorage.	Depends on ASS-STORAGE
FR-010	Functional	System MUST load expenses from localStorage on application start.	-
FR-011	Functional	System MUST show an empty state when no expenses exist.	-
FR-012	Functional	System MUST update totals in real-time when expenses are added or deleted.	Implements US-TOTALS
FR-013	Constraint	System MUST NOT allow editing/updating existing expenses in v1.	Out of Scope
FR-014	Functional	System MUST display most recent 50 expenses by default with 'Load more'.	-
ENT-EXPENSE	Entity	Expense: id, amount, date, category, description, createdAt.	-
ENT-CATEGORY	Entity	Category: Predefined list (Food, Transport, Entertainment, etc.).	-
SC-001	Success	Users can add an expense in under 30 seconds from dashboard load.	-
SC-004	Success	Dashboard totals are always accurate and match the sum of expenses.	-
SC-005	Success	Data persists across browser sessions.	Relates to FR-009

ASS-AUTH	Assumption	Single-user personal tracker - no authentication, no multi-user support.	-
ASS-STORAGE	Assumption	Data stored locally in browser localStorage - no backend, no cloud sync.	-

3.2 Security Rules

• **Data Isolation**: No server-side storage; data is isolated to the user's local browser profile.

• **Input Validation**: Strict positive-number validation for monetary amounts to prevent negative balance injection.

3.3 Data Models

• **Expense (ENT-EXPENSE)**:

• `id`: Unique String (Primary Key)

• `amount`: Decimal

• `date`: ISO String

• `category`: String (Foreign Key to ENT-CATEGORY)

• `description`: String (Optional)

• `createdAt`: Timestamp

• **Category (ENT-CATEGORY)**:

• `category_name`: String (Unique identifier)

4. Project Governance & Structural Gaps

4.1 Structural Gaps

Gap	Priority	Remediation Advice
Scope & Out-of-Scope	MEDIUM	Create a dedicated section for Scope. Although FR-013 mentions out-of-scope items, a formal boundary list is needed.

4.2 Remediation & Workflow

The identified gap regarding the "Scope" section should be addressed by consolidating all "MUST NOT" requirements (like FR-013) and assumptions (ASS-AUTH, ASS-STORAGE) into a single boundary document to prevent scope creep in future versions.

5. Technical & Domain Glossary (Terminology Reference)

Term	Category	Context Anchor	Project Definition

Category	BUSINESS_DOMAIN	ENT-CATEGORY	A predefined set of labels including Food, Transport, Entertainment, Shopping, Utilities, and Other used to classify spending.
Expense	BUSINESS_DOMAIN	ENT-EXPENSE	A financial record comprising a unique identifier, decimal value, ISO date, classification label, optional text, and a creation timestamp.
Fixed-Point Numeric Constraint	TECHNICAL_STACK	FR-002	A validation rule ensuring the monetary value is a positive number strictly greater than zero.
LocalStorage	TECHNICAL_STACK	FR-009	The browser-based key-value storage mechanism used for persisting data across sessions without a backend.
NOT	TECHNICAL_STACK	FR-013	A logical negation used to explicitly define out-of-scope operations, specifically forbidding the modification of existing records in the initial version.